

DART Installation Guide

This document provides detailed installation instructions for DART (Digital Asset Routing and Transformation) on macOS, Windows, and Linux systems.

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macOS Installation (DMG)

DART for macOS is distributed as a DMG file with an automated installer that requires **no admin permissions**.

Prerequisites

- **macOS 12** (Monterey) or later
- **Python 3.8+** must be installed
 - Download from: python.org/downloads
 - Or install via Homebrew: `brew install python`
 - Verify: `python3 --version`

Installation Steps

1. Download the DMG

- Download `DART_v2.x.dmg` from your distribution source

2. Open the DMG

- Double-click the downloaded DMG file
- A new window opens showing the installer contents

3. Run the Installer

- Double-click: `Install DART.command`

Note for first-time users (unsigned app):

- If the installer won't open, macOS Gatekeeper is blocking it
- Right-click on `Install DART.command` → select "Open"
- Click "Open" in the security dialog

- This is only needed once

4. Follow the Installation

- A Terminal window opens with the installer
- The installer checks for Python 3
- Confirms installation location: `~/DART/`
- Copies all files (takes a few seconds)
- Creates a launcher on your Desktop
- Press Enter when finished

5. Eject the DMG

- Right-click the mounted DART volume → Eject
- Or drag it to the Trash

Installation Location

DART is installed to: `~/DART/` (your home directory)

```
~/DART/  
├─ Launch DART.command    # Double-click to start DART  
├─ app.py                 # Main application  
├─ run.sh                 # Launcher script  
├─ common_dg_utilities/   # Utility functions (included)  
├─ INSTALLATION.md        # This file  
├─ README.md              # Full documentation  
└─ ... (other files)
```

Desktop Shortcut

The installer creates: **Launch DART** on your Desktop

- Double-click this to launch DART at any time
- It's a symbolic link to `~/DART/Launch DART.command`

First Launch

1. Start DART

- Double-click: `Launch DART` (on Desktop)
- Or navigate to `~/DART/` and double-click: `Launch DART.command`

2. Automatic Setup (first time only)

- A Terminal window opens
- Python virtual environment is created in `~/DART/.venv/`
- Dependencies install automatically (1-2 minutes)
- Progress is shown in the Terminal

3. DART Window Opens

- The DART application window appears when ready
- The Terminal window stays open (don't close it!)
- Minimize the Terminal if you want it out of the way

4. Subsequent Launches

- Just double-click the launcher
- DART starts immediately (no setup needed)
- Dependencies are already installed

Why No Applications Folder?

Traditional approach issues:

- Many work computers restrict Applications folder access
- Requires admin permissions to install
- Users encounter permission errors

DART's solution:

- Installs to `~/DART/` in your home directory
 - No admin permissions required
 - Works on all systems, including managed/restricted computers
 - Clean, predictable location
-

Windows Installation (ZIP)

DART for Windows is distributed as a ZIP archive.

Prerequisites

- **Windows 10** or **Windows 11**
- **Python 3.8+** must be installed
 - Download from: python.org/downloads
 - **During installation: Check "Add Python to PATH"** 
 - Verify: Open Command Prompt, type `python --version`

Installation Steps

1. Download the ZIP

- Download `DART_v2.x_Windows.zip` from your distribution source

2. Extract the Archive

- Right-click the ZIP file → Extract All...
- Choose a convenient location (e.g., `C:\Users\YourName\DART\`)
- Or extract to your Desktop

3. Installation Location

- The extracted folder contains all DART files
- No further installation needed

First Launch

1. Open the DART folder

- Navigate to where you extracted the files

2. Run the launcher

- Double-click: `run.bat`

3. Automatic Setup (first time only)

- A Command Prompt window opens
- Python virtual environment is created in `.venv\`
- Dependencies install automatically (1-2 minutes)
- Progress is shown in the console

4. DART Window Opens

- The DART application window appears when ready
- The Command Prompt window stays open (don't close it!)
- Minimize the console if you want it out of the way

5. Subsequent Launches

- Just double-click `run.bat`
- DART starts immediately

Creating a Desktop Shortcut (Optional)

1. Right-click `run.bat` → Send to → Desktop (create shortcut)
2. Rename the shortcut to "DART"
3. Double-click the Desktop shortcut to launch DART

Linux Installation (Source)

DART runs from source on Linux systems.

Prerequisites

- **Python 3.8+**

- Most Linux distributions include Python 3
- Verify: `python3 --version`
- Install if needed:
 - Debian/Ubuntu: `sudo apt install python3 python3-venv`
 - Fedora/RHEL: `sudo dnf install python3`

- Arch: `sudo pacman -S python`

Installation Steps

1. Download or Clone

```
# If you have the source archive:
unzip DART_v2.x.zip
cd DART_v2.x

# Or clone from repository:
git clone https://github.com/Digital-Grinnell/DART.git
cd DART
```

2. Run DART

```
./run.sh
```

3. First Launch

- The script creates a Python virtual environment in `.venv/`
- Installs all dependencies automatically
- Launches the DART application
- Subsequent runs skip the setup phase

Creating a Desktop Launcher (Optional)

Create `~/.local/share/applications/dart.desktop`:

```
[Desktop Entry]
Type=Application
Name=DART
Comment=Digital Asset Routing and Transformation
Exec=/path/to/DART/run.sh
Icon=utilities-terminal
Terminal=false
Categories=Utility;
```

Update `/path/to/DART/` with your actual path.

Prerequisites

Python 3.8 or Later

DART requires Python 3.8 or later. To verify your Python version:

```
# macOS/Linux
python3 --version

# Windows
python --version
```

Installation:

- **macOS:**
 - Download: python.org/downloads
 - Or via Homebrew: `brew install python`
- **Windows:**
 - Download: python.org/downloads
 - **Important:** Check "Add Python to PATH" during installation
- **Linux:**
 - Usually pre-installed
 - If needed: `sudo apt install python3` (Debian/Ubuntu)

Python Dependencies

The following Python packages are installed automatically on first launch:

- **flet 0.25.2** — UI framework
- **flet-desktop 0.25.2** — Desktop-specific features
- **cryptography** — Encrypted settings storage
- **azure-storage-blob 12.19.0+** — Azure Blob Storage integration
- **Pillow** — Image processing
- **PyMuPDF (fitz)** — PDF processing
- **pandas** — Data manipulation

You don't need to install these manually — the `run.sh` / `run.bat` scripts handle everything.

Troubleshooting

macOS Issues

"Install DART.command can't be opened"

Cause: macOS Gatekeeper blocking unsigned app

Solution:

1. Right-click on `Install DART.command`
2. Select "Open" from the menu

3. Click "Open" in the security dialog
4. Only needed once

"Python 3 not found" error

Cause: Python not installed or not in PATH

Solution:

```
# Install Python via Homebrew
brew install python

# Or download from python.org
# Then verify:
python3 --version
```

Installer says "DART is already installed"

Cause: Previous DART installation exists

Solution:

- Type **y** to overwrite the old installation
- Or type **n** to cancel and manually remove **~/DART/** first

DART window doesn't appear

Cause: Terminal window was closed during setup

Solution:

1. Open Terminal
2. Navigate: **cd ~/DART**
3. Run manually: **bash run.sh**
4. Watch for error messages

Windows Issues

"python is not recognized"

Cause: Python not installed or not in PATH

Solution:

1. Install Python from python.org
2. **During installation: Check "Add Python to PATH"**
3. Restart Command Prompt
4. Verify: **python --version**

Dependencies fail to install

Cause: Network issues or pip outdated

Solution:

```
# Upgrade pip
python -m pip install --upgrade pip

# Then re-run DART
run.bat
```

Virtual environment errors

Cause: Corrupt virtual environment

Solution:

```
# Delete the virtual environment
rmdir /s .venv

# Re-run launcher (recreates .venv)
run.bat
```

Linux Issues

./run.sh: Permission denied

Cause: Script not executable

Solution:

```
chmod +x run.sh
./run.sh
```

"No module named 'tkinter'"

Cause: tkinter not installed (required by flet)

Solution:

```
# Debian/Ubuntu
sudo apt install python3-tk

# Fedora/RHEL
sudo dnf install python3-tkinter
```


Common Issues (All Platforms)

DART closes immediately after launch

Cause: Error during startup

Solution:

1. Look for error messages in the Terminal/Console window
2. Check log files in `~/DART-data/logfiles/` or `{working_folder}/logfiles/`
3. Common causes:
 - Missing Python dependencies (re-run installer)
 - Corrupted settings file (delete `~/DART-data/persistent.json`)

"common_dg_utilities not found" error

Cause: DMG/ZIP was missing the utilities folder

Solution:

- This should not happen with properly built installers
- Contact support or rebuild DMG/ZIP with utilities included
- For developers: Ensure `../common-DG-utilities/` exists when building

Uninstallation

macOS

1. Delete the application folder:

```
rm -rf ~/DART
```

2. Delete user data (optional):

```
rm -rf ~/DART-data
```

3. Remove Desktop shortcut:

- Drag **Launch DART** from Desktop to Trash

Windows

1. Delete the DART folder:

- Navigate to where you extracted DART
- Delete the entire folder

2. Delete user data (optional):

- Navigate to: `C:\Users\YourName\DART-data\`
- Delete the folder

3. Remove Desktop shortcut (if created):

- Delete the DART shortcut from Desktop

Linux

```
# Delete DART folder
rm -rf ~/path/to/DART

# Delete user data (optional)
rm -rf ~/DART-data

# Remove desktop launcher (if created)
rm ~/.local/share/applications/dart.desktop
```

Support

Documentation

- **README.md** — Full application documentation
- **QUICKSTART.md** — Quick reference guide
- **Function Help** — Built-in help for each function (Help Mode toggle in app)

Getting Help

- **Repository:** <https://github.com/Digital-Grinnell/DART>
- **Issues:** Report bugs on GitHub Issues
- **Questions:** Use GitHub Discussions

Log Files

DART maintains detailed log files for troubleshooting:

- **Startup logs:** `~/DART-data/logfiles/`
- **Operational logs:** `{working_folder}/logfiles/` (set in app)
- **Format:** `dart_YYYYMMDD_HHMMSS.log`

Include relevant log excerpts when reporting issues.

System Requirements

Minimum

- **OS:** macOS 12+, Windows 10+, or modern Linux
- **Python:** 3.8 or later
- **Memory:** 2 GB RAM
- **Disk:** 500 MB free space
- **Display:** 1280x720 or higher

Recommended

- **OS:** macOS 13+, Windows 11, or latest LTS Linux
- **Python:** 3.10 or later
- **Memory:** 4 GB RAM
- **Disk:** 1 GB free space
- **Display:** 1920x1080 or higher

Security Note

DART stores sensitive settings (like Azure connection strings) using encrypted storage:

- **Encryption key:** `~/DART-data/encryption_key`
- **Settings file:** `~/DART-data/persistent.json` (sensitive fields encrypted)
- **Working settings:** `{working_folder}/dart_settings.json` (per-project)

Keep your encryption key secure. If lost, you'll need to re-enter encrypted settings.

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DART Version: 2.x

Documentation: <https://github.com/Digital-Grinnell/DART>